



MT1008-Multivariable Calculus

Program: BS(CS)

Credit Hours: 03

Instructor: Dr. Askar Ali

Semester: Fall 2024

Pre-requisite: MT1003

Email: askar.ali@nu.edu.pk

Course Description:

The main objective of the course is to make students proficient in techniques and skills of multivariable calculus. The emphasis will be on application of learned techniques to the solution of problems of science and engineering.

Text & Reference Books:

Text: “Calculus: Early Transcendentals” by James Stewart, 9th Edition Brooks / Cole USA.

Reference: “Thomas’ Calculus” by George B. Thomas, Jr., Maurice D. Weir, Joel R. Hass. 15th Edition, Pearson, USA.

Assessment Plan:

Quizzes	10%
Homework assignments	10%
Sessional I	15%
Sessional II	15%
Final exam	50%

Course Learning Outcomes:

Upon successful completion of this course, students will be able to:

1. Comprehend partial derivatives, extremas and chain rule for multivariable functions.
2. Evaluate multiple integrals and discuss their applications.
3. Solve problems involving line integral and surface integral.

Contents:

Vector equation of a line, various forms of equation of a line. Vector and scalar equation of a plane	12.5
Traces, cylinders, and quadric surfaces	12.6
Introduction of plane and space curves and $\mathbf{r}(t)$, helix	13.1
Derivative of $\mathbf{r}(t)$	13.2
Arc length of parametric curves, Arc length function, ds derivation, unit tangent definition	13.3
Functions of several variables, domain, graph, level curves, contour map, level surfaces	14.1
Limit, and continuity of functions of several (two, three and more) variables, composition of function	14.2
Partial derivatives, its interpretation, higher derivatives, Laplace, and wave equations.	14.3
Tangent planes and linear approximations, differentials	14.4
Chain rule of differentiation of function of two and three variables, implicit differentiation.	14.5
Directional derivatives and the gradient vector	14.6
Double integrals over rectangles, volume, average value	15.1
Double Integrals over General Regions (Type I, Type II etc.), Changing the order of integration, $A(D)$ using double integral	15.2
Double Integrals in Polar Coordinates, Jacobian	15.3
Surface area of $z = f(x, y)$	15.5
Triple Integrals (x -simple, y -simple, z -simple, projections)	15.6
Definition of vector field in R^2 and R^3 , Recall gradient of a scalar function, definition of conservative vector field	16.1

Parametrization of well-known curves, Line integral of $f(x,y)$ over a plane curve C with respect to s , x and y respectively, line integral of $f(x,y,z)$ over a space curve C with respect to s , x , y and z respectively, line integrals of vector fields, Work	16.2
The fundamental theorem for line integrals, work done, conservative vector fields and potential functions	16.3
Simple curve, closed curve, simple closed curve, connected region, simply connected region, positive orientation, negative orientation, Green's Theorem	16.4
Curl and divergence of a vector field, theorem for checking whether a field is conservative ? irrotational vector field, incompressible vector field	16.5
Parametric surfaces and their areas, Surface integrals.	16.6, 16.7
Stokes' Theorem, The Divergence Theorem	16.8, 16.9